CHMENG 250: Transport Processes

Homework 3: Stress, Momentum Balances, and Steady Viscous Flows

Assigned: Fri, Sep. 18, 9:00 a.m. (PT) **Due:** Sat, Oct. 3, 11:59 p.m. (PT) **Total:** 100 points

Scope and conventions. This assignment covers Chapter 3 and the steady-flow material of Chapter 4 through circular Couette flow, together with the gravity-driven-film boundary conditions. All fields are as smooth as the operations performed on them require, repeated indices are summed, and

$$L_{ij} = \partial_j v_i, \qquad D_{ij} = \tfrac{1}{2}\left(L_{ij} + L_{ji}\right), \qquad t_i = T_{ij} n_j, \qquad (\boldsymbol{\nabla}\cdot\boldsymbol{T})_i = \partial_j T_{ij}.$$

Unless a wall-to-fluid normal is explicitly specified, $\boldsymbol{n}$ is outward from the fluid, so $\boldsymbol{T}\boldsymbol{n}$ is the traction exerted on that fluid by its surroundings. You may use the Reynolds transport theorem corollary derived in Lecture 7, the differential identities proved in Homework 2, and the coordinate formulas of Appendix 3.C without rederiving them. Show the reduction and the boundary conditions before using a velocity profile, and include units and force directions in any physical interpretation.

Problem 1: Action and reaction across an imagined cut [10 points]

How can a force be internal to a body but external to one of its parts? Let a material body $V(t)$ be divided into two subbodies $V_1(t)$ and $V_2(t)$ with disjoint interiors and a common smooth cut $\Sigma(t)$. The cut is imagined within a continuous material: it carries no separate mass and no surface stress. The outward normals on its two sides are $\boldsymbol{n}$ and $-\boldsymbol{n}$. Assume Cauchy's hypothesis $\boldsymbol{t} = \boldsymbol{t}(\boldsymbol{x}, t; \boldsymbol{n})$ and that the traction is continuous in position for fixed t and $\boldsymbol{n}$, but do not assume the stress theorem $\boldsymbol{t} = \boldsymbol{T}\boldsymbol{n}$ yet.

(a) Sketch the body and the two separated subbodies. At one pair of matching points label the outward normals and the forces "V_2 on V_1" and "V_1 on V_2." Explain what Cauchy's hypothesis says about two different surfaces sharing the same normal at that point. [2 points]

(b) Write the linear-momentum balance for each subbody, separating its body force, the traction on the inherited exterior boundary, and the traction on Σ. Add the two balances and subtract the balance for the whole body to show that

$$\int_{\Sigma} \left[\boldsymbol{t}(\boldsymbol{n}) + \boldsymbol{t}(-\boldsymbol{n})\right] \mathrm{d}A = \boldsymbol{0}. \qquad \text{[3 points]}$$

(c) Explain why repeating the construction on arbitrary local patches gives $\boldsymbol{t}(-\boldsymbol{n}) = -\boldsymbol{t}(\boldsymbol{n})$. Identify the regularity assumption used in this localization step. [3 points]

(d) Once a linear representation $\boldsymbol{t} = \boldsymbol{T}\boldsymbol{n}$ has been established, does action and reaction alone imply $\boldsymbol{T} = \boldsymbol{T}^{\top}$? Give a concrete nonsymmetric 3×3 matrix that settles the question, and identify the additional balance law that will rule it out for a classical nonpolar continuum. [2 points]

Problem 2: Two deformation modes and the Newtonian law [12 points]

A compressible Newtonian fluid has

$$\boldsymbol{T} = -p\boldsymbol{I} + \boldsymbol{\tau}, \qquad \boldsymbol{\tau} = 2\mu\boldsymbol{D}' + \eta\theta\boldsymbol{I}, \qquad \boldsymbol{D}' = \boldsymbol{D} - \tfrac{1}{3}\theta\boldsymbol{I}, \qquad \theta = \operatorname{tr}\boldsymbol{D},$$

where p is the thermodynamic pressure, μ is the constant shear viscosity, and η is the constant bulk viscosity. This problem concerns the local constitutive law alone; you are not asked to solve a compressible flow or an energy equation.

(a) Express the same law as $\boldsymbol{\tau} = 2\mu\boldsymbol{D} + \lambda\theta\boldsymbol{I}$, giving λ in terms of μ and η. Find $\operatorname{tr}\boldsymbol{T}$ and the mean compressive normal stress $p_m = -\operatorname{tr}\boldsymbol{T}/3$. Under what condition does $p_m = p$? [3 points]

(b) Using $D_{ij} = (\partial_j v_i + \partial_i v_j)/2$, calculate $\partial_j T_{ij}$. Obtain the constant-viscosity momentum equation and state its incompressible reduction. Is Stokes' hypothesis $\eta = 0$ needed for that reduction? [3 points]

(c) Show that the viscous dissipation is

$$\Phi = \boldsymbol{\tau} : \boldsymbol{D} = 2\mu\, \boldsymbol{D}' : \boldsymbol{D}' + \eta\theta^2.$$

Use a traceless deformation and a pure uniform dilation to show that $\Phi \geq 0$ for every $\boldsymbol{D}$ requires, and is guaranteed by, $\mu \geq 0$ and $\eta \geq 0$. [3 points]

(d) For uniform dilation $\boldsymbol{v} = a\boldsymbol{x}$, where $a > 0$ is constant, and simple shear $\boldsymbol{v} = \dot{\gamma} y\, \boldsymbol{e}_x$, where $\dot{\gamma} > 0$ is the constant shear rate, compute $\boldsymbol{\tau}$ and Φ. Which viscosity can each motion probe? [3 points]

Problem 3: Where does the mechanical work go? [12 points]

Consider an incompressible Newtonian fluid with velocity field $\boldsymbol{v}$, constant density ρ, shear viscosity $\mu > 0$, and symmetric stress tensor $\boldsymbol{T} = -p\boldsymbol{I} + 2\mu\boldsymbol{D}$. You may use the identity proved in Homework 2,

$$\boldsymbol{v} \cdot (\boldsymbol{\nabla} \cdot \boldsymbol{T}) = \boldsymbol{\nabla} \cdot \left(\boldsymbol{T}^{\top}\boldsymbol{v}\right) - \boldsymbol{T} : \boldsymbol{\nabla}\boldsymbol{v}, \qquad \boldsymbol{T} : \boldsymbol{\nabla}\boldsymbol{v} = \boldsymbol{T} : \boldsymbol{D},$$

(a) Take the dot product of the momentum equation with $\boldsymbol{v}$ and derive the local kinetic-energy balance

$$\rho \frac{\mathrm{D}}{\mathrm{D}t}\left(\frac{|\boldsymbol{v}|^2}{2}\right) = \boldsymbol{\nabla} \cdot (\boldsymbol{T}\boldsymbol{v}) + \rho\, \boldsymbol{b} \cdot \boldsymbol{v} - 2\mu\, \boldsymbol{D} : \boldsymbol{D},$$

where $\boldsymbol{b}$ is the body force per unit mass of the fluid. Does the disappearance of $p\, \boldsymbol{\nabla} \cdot \boldsymbol{v}$ mean that pressure cannot transfer mechanical energy? [3 points]

(b) Integrate over a fixed domain with a stationary no-slip boundary and $\boldsymbol{b} = \boldsymbol{0}$. Show that the total kinetic energy cannot increase. [3 points]

(c) Now let the upper plate of a plane Couette cell move at speed U, with the lower plate fixed and gap H. For the steady profile $u = Uy/H$, compute the power supplied by the upper plate per unit

area and the dissipation integrated across the gap. Verify their equality and state the direction of the plate-on-fluid tangential traction. [3 points]

(d) At one point in a Newtonian fluid, $\boldsymbol{D} = \mathrm{diag}(a, -a, 0)$, where $a \neq 0$ is a constant rate. Using the viscous dissipation expression established in Problem 2(c), find Φ and explain why zero volume change does not mean zero dissipation. [3 points]

Problem 4: Pressure and traction at a fluid interface [10 points]

An interface separates two Newtonian fluids. Let $\boldsymbol{n}$ point from Fluid 1 into Fluid 2, and let $\boldsymbol{s}$ be a unit tangent. Neglect interfacial mass and inertia, and take the surface tension γ as the only surface stress. You may use the balance stated in Chapter 3,

$$\left(\boldsymbol{T}^{(2)} - \boldsymbol{T}^{(1)}\right)\boldsymbol{n} = \gamma\kappa\boldsymbol{n} - \boldsymbol{\nabla}_s\gamma, \qquad \kappa = \boldsymbol{\nabla}_s \cdot \boldsymbol{n}.$$

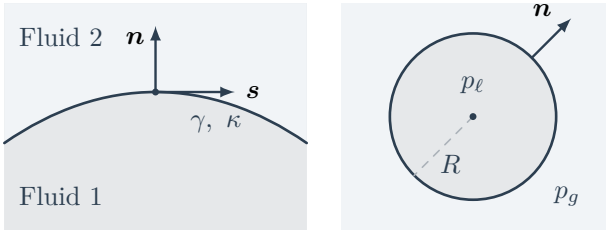


Figure 1: Traction at a fluid interface. **(a)** The unit normal $\boldsymbol{n}$ points from Fluid 1 into Fluid 2, while $\boldsymbol{s}$ is a unit tangent to the interface; γ and κ denote the surface tension and curvature. **(b)** A spherical liquid drop of radius R and internal pressure p_ℓ, at rest in a gas at pressure p_g.

(a) Project this equation onto $\boldsymbol{n}$ and $\boldsymbol{s}$, and explain why a curved interface need not have continuous normal traction. [2 points]

(b) A spherical liquid drop of radius R rests in a gas at pressure p_g, with constant γ. Find the liquid pressure. Which side has the higher pressure, and why does this not contradict the action–reaction result of Problem 1? [2 points]

(c) At a flat liquid–gas interface, let Fluid 1 be the liquid and Fluid 2 the gas, with $\boldsymbol{T}^{(2)} = -p_g\boldsymbol{I}$ and $\boldsymbol{T}^{(1)} = -p\boldsymbol{I} + 2\mu\boldsymbol{D}$, and take γ to be constant. Derive the conditions on p and on $\boldsymbol{s} \cdot \boldsymbol{D}\boldsymbol{n}$. Does the normal-stress balance necessarily imply $p = p_g$? Treat this as a local boundary calculation, without assuming a complete flow field. [3 points]

(d) In two dimensions (2D), a material interface is the graph $y = h(x, t)$, with no phase change or mass transfer. Use $F(x, y, t) = y - h(x, t)$ and $\mathrm{D}F/\mathrm{D}t = 0$ to derive its kinematic condition. Explain how a flat interface can stay at a fixed height while the fluid on it moves tangentially. [3 points]

Problem 5: Frame indifference and why the stress cannot feel the spin [14 points]

[Advanced] The Newtonian constitutive law of Chapter 3 is built on the rate of strain $\boldsymbol{D}$ and never on the spin $\boldsymbol{W}$. This problem shows that the omission is forced rather than chosen. Two observers assign to the same event the positions

$$\boldsymbol{x}^* = \boldsymbol{Q}(t)\,\boldsymbol{x} + \boldsymbol{c}(t), \qquad \boldsymbol{Q}^{\top}\boldsymbol{Q} = \boldsymbol{I},$$

with $\boldsymbol{Q}(t)$ a rotation and $\boldsymbol{c}(t)$ a translation, both possibly time dependent. A tensor field $\boldsymbol{A}$ is called *objective* when the two observers' descriptions are related by $\boldsymbol{A}^* = \boldsymbol{Q}\boldsymbol{A}\boldsymbol{Q}^{\top}$, that is, when $\boldsymbol{A}$ merely co-rotates with the observer. Frame indifference is the requirement that the *material response* be the same for both observers; it is not a claim that every field they measure agrees.

(a) Differentiate the position relation along one material particle to obtain $\boldsymbol{v}^*$ in terms of $\boldsymbol{v}$, $\boldsymbol{x}$, $\boldsymbol{Q}$, and $\boldsymbol{c}$. Hence explain why the velocity is *not* objective, and identify physically the terms responsible. [2 points]

(b) Let $\boldsymbol{L} = \boldsymbol{\nabla v}$ be the velocity-gradient tensor, so that the starred observer forms $L^*_{ij} = \partial v^*_i / \partial x^*_j$. Using $\boldsymbol{x} = \boldsymbol{Q}^{\top}\,(\boldsymbol{x}^* - \boldsymbol{c})$ and the chain rule, show that

$$\boldsymbol{L}^* = \boldsymbol{Q}\,\boldsymbol{L}\,\boldsymbol{Q}^{\top} + \dot{\boldsymbol{Q}}\,\boldsymbol{Q}^{\top}$$ [3 points]

Hint

Both $\boldsymbol{Q}$ and $\boldsymbol{c}$ depend on t alone, so differentiating $v^*_i = \dot{Q}_{ik}x_k + Q_{ik}v_k + \dot{c}_i$ with respect to x_m is straightforward; then convert $\partial/\partial x_m$ into $\partial/\partial x^*_j$.

(c) Differentiate $\boldsymbol{Q}\boldsymbol{Q}^{\top} = \boldsymbol{I}$ to show that $\dot{\boldsymbol{Q}}\boldsymbol{Q}^{\top}$ is antisymmetric. Split $\boldsymbol{L} = \boldsymbol{D} + \boldsymbol{W}$ into symmetric and antisymmetric parts and read off

$$\boldsymbol{D}^* = \boldsymbol{Q}\,\boldsymbol{D}\,\boldsymbol{Q}^{\top}, \qquad \boldsymbol{W}^* = \boldsymbol{Q}\,\boldsymbol{W}\,\boldsymbol{Q}^{\top} + \dot{\boldsymbol{Q}}\,\boldsymbol{Q}^{\top}.$$

Which of the two is objective, and what is the physical meaning of the extra term? [2 points]

(d) Suppose the viscous stress at a point is determined by the velocity gradient through a constitutive function,

$$\boldsymbol{\tau} = \mathcal{F}\,(\boldsymbol{D}, \boldsymbol{W})\,,$$

and that the response is frame indifferent, meaning $\mathcal{F}\,(\boldsymbol{D}^*, \boldsymbol{W}^*) = \boldsymbol{Q}\,\mathcal{F}\,(\boldsymbol{D}, \boldsymbol{W})\,\boldsymbol{Q}^{\top}$ for every admissible observer change. Fix the event under consideration and an instant t_0, and let $\boldsymbol{W}_0$ denote the value of $\boldsymbol{W}$ there at t_0. Choose the rotation

$$\boldsymbol{Q}(t) = \exp\left[-\,(t - t_0)\,\boldsymbol{W}_0\right], \qquad \text{so that} \qquad \boldsymbol{Q}(t_0) = \boldsymbol{I}, \quad \dot{\boldsymbol{Q}}(t_0) = -\boldsymbol{W}_0.$$

Verify that this $\boldsymbol{Q}$ is an admissible rotation, evaluate $\boldsymbol{D}^*$ and $\boldsymbol{W}^*$ at t_0, and deduce that

$$\mathcal{F}(\boldsymbol{D}, \boldsymbol{W}) = \mathcal{F}(\boldsymbol{D}, \boldsymbol{0}),$$

for every $\boldsymbol{D}$ and every $\boldsymbol{W}$. State in one sentence what this proves about the Newtonian law. [4 points]

Hint

$\boldsymbol{W}_0$ is antisymmetric, so $-\boldsymbol{W}_0$ is an admissible angular velocity; the exponential of an antisymmetric tensor is a rotation. At t_0 the first argument is untouched while the second is driven to zero.

(e) Explain why the conclusion of part (d) does *not* say that the local rotation is physically irrelevant. Then check the result against rigid-body rotation $\boldsymbol{v} = \boldsymbol{\omega} \times \boldsymbol{x}$ with constant angular-velocity vector $\boldsymbol{\omega}$: compute $\boldsymbol{D}$ and $\boldsymbol{W}$, and state what your answer requires of the viscous stress relative to its zero-deformation value. [3 points]

Problem 6: Torque between rotating cylinders [14 points]

Adapted from Fall 2022 PS3, Problem 2, with the circular-flow reduction of Chapter 4. An incompressible Newtonian fluid with constant density ρ and shear viscosity $\mu > 0$ fills the annulus $R_i < r < R_o$ between long coaxial cylinders. The cylinders rotate at constant angular velocities $\Omega_i \geq \Omega_o \geq 0$. Neglect gravity and end effects, and seek the steady laminar field $\boldsymbol{v} = v_\theta(r)\boldsymbol{e}_\theta$, $p = p(r)$. You may use the cylindrical operators of Appendix 3.C, including

$$D_{r\theta} = D_{\theta r} = \tfrac{1}{2}\left(v_\theta' - v_\theta/r\right), \qquad \omega_z = \frac{1}{r}\frac{\mathrm{d}\left(r v_\theta\right)}{\mathrm{d}r}.$$

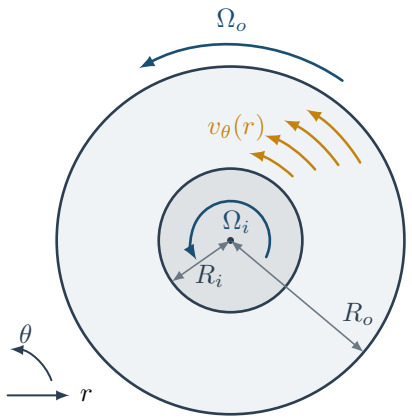


Figure 2: Cross-section of the fluid annulus $R_i < r < R_o$ between long coaxial cylinders rotating at angular velocities Ω_i and Ω_o. The curved arrows indicate the azimuthal velocity $v_\theta(r)$ schematically; their lengths are not proportional to the magnitude of $v_\theta = Ar + B/r$.

(a) Verify continuity, calculate the material acceleration, and state the no-slip conditions. Is zero convective acceleration justified simply because the field is steady and axisymmetric? [2 points]

(b) Reduce the azimuthal momentum equation and obtain $v_\theta = Ar + B/r$. Determine A and B from the two cylinder speeds. Check the special cases $\Omega_o = 0$ and $\Omega_i = \Omega_o$. [4 points]

(c) Use radial momentum and $p(R_o) = p_o$ to express $p(r)$ as a definite integral of the known velocity. You need not evaluate the integral. [2 points]

(d) Find $T_{r\theta}(r)$ and the signed axial torque exerted by the fluid on an inner-cylinder length L. Hence find the external torque needed to maintain Ω_i, and state its sign when $\Omega_i > \Omega_o$. Explain why the magnitude of the transmitted torque is independent of radius. [3 points]

(e) Find the vorticity and the viscous dissipation $\Phi(r)$. For solid-body rotation, what becomes of the torque, the dissipation, and the radial pressure gradient? [3 points]

Problem 7: Pressure-driven flow and the force on a pipe [14 points]

Adapted from Fall 2022 PS4, Problem 1. A Newtonian fluid with constant density ρ and shear viscosity $\mu > 0$ flows steadily through a straight circular pipe of radius R. Neglect gravity and entrance effects. Over an axial length L the pressure drop is $\Delta p = p(0) - p(L) > 0$. Start from the axisymmetric form $\boldsymbol{v} = u(r, z)\boldsymbol{e}_z$, with no slip at $r = R$ and regularity at $r = 0$.

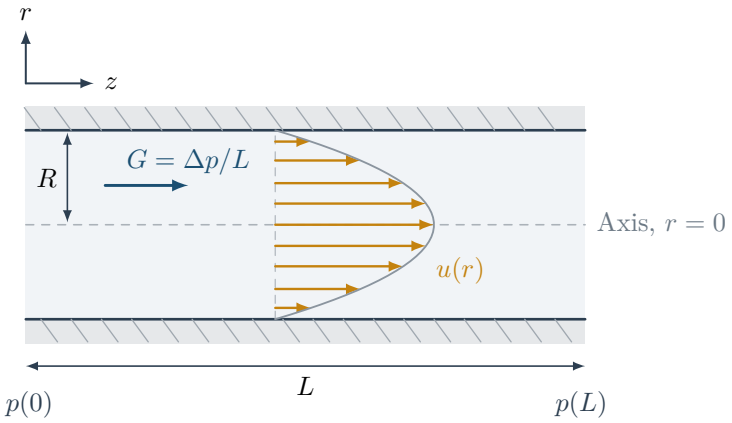


Figure 3: Longitudinal section through the axis of a circular pipe of radius R. The pressure falls by Δp over the axial length L, driving the parabolic velocity profile $u(r)$. The velocity arrows indicate the shape and direction of the flow schematically; their lengths are not drawn in proportion to the local speed.

(a) Use continuity to show that u is independent of z. Evaluate the convective acceleration. Does this reduction require a small Reynolds number, or is it exact for the assumed field? [2 points]

(b) Use transverse momentum to determine the cross-sectional dependence of the pressure, and show that the axial pressure gradient is constant. With $G = \Delta p/L$, solve

$$\frac{\mu}{r}\frac{\mathrm{d}}{\mathrm{d}r}\left(r\frac{\mathrm{d}u}{\mathrm{d}r}\right) = -G.$$ [4 points]

(c) Calculate the volume flow rate Q, the mean speed $\bar{u}$, and the centerline speed. At fixed Δp, L, and μ, by what factor does Q change if R is reduced by 10%? At fixed Q, by what factor must Δp

change instead? [3 points]

(d) Find $T_{zr}(r)$ and the axial force exerted by the wall on the fluid over the length L. Check the global axial momentum balance, and give the opposite force exerted by the fluid on the pipe wall. [3 points]

(e) For this flow $\Phi = \mu\,(\mathrm{d}u/\mathrm{d}r)^2$. Integrate the dissipation over the length L and show that it equals $Q\Delta p$. Why can the stationary wall exert a drag force while doing no work? [2 points]

Problem 8: Can a rising belt carry a draining film upward? [14 points]

Adapted from Fall 2022 PS4, Problem 4, using the wall and free-surface conditions of Chapters 3 and 4. A vertical belt moves upward at speed $U > 0$ and carries a Newtonian liquid film of uniform prescribed thickness h. Let x point upward along the belt and y from the belt into the liquid, with the belt at $y = 0$ and the free surface at $y = h$. The density ρ and shear viscosity $\mu > 0$ are constant, and $\boldsymbol{g} = -g\boldsymbol{e}_x$. The gas has uniform pressure p_a and negligible viscosity, and the surface tension is uniform. Seek a steady fully developed flow $\boldsymbol{v} = u(y)\boldsymbol{e}_x$ away from the bath and the belt ends. You are not asked to determine how the coating thickness is selected.

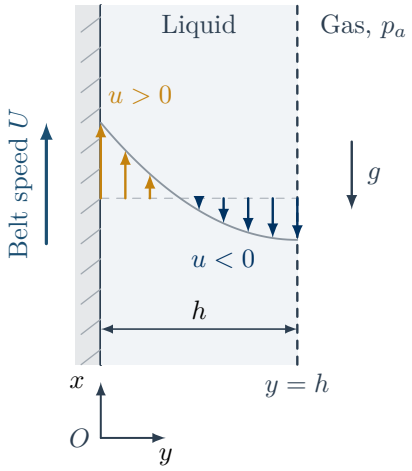


Figure 4: A Newtonian film of uniform thickness h carried by a belt rising at speed U, with x up along the belt and y measured from the belt into the liquid. The profile drawn is the intermediate case of part (c): the liquid near the belt rises while an outer layer still drains, so that the net flux may be upward even though part of the film moves down.

(a) State the wall and free-surface boundary conditions on velocity and traction. Use normal momentum and the normal-stress balance to establish the pressure field; do not assume a streamwise pressure gradient that cancels gravity. [2 points]

(b) Reduce the streamwise momentum equation, solve for $u(y)$, and sketch the profile for one speed high enough that the entire film moves upward. [4 points]

(c) Compute the upward volume flux per unit belt width, $q = \int_0^h u\,\mathrm{d}y$. Find the critical belt speed above which the net flux is upward and, separately, above which every point in the film moves upward. Can there be net upward transport while part of the film drains downward? [3 points]

(d) Find the tangential traction exerted by the belt on the liquid and the opposite traction on the belt. Verify that the upward belt force per unit area balances the weight of the film per unit area, even when the net flux is zero. [3 points]

(e) Evaluate the dissipation integrated across the film and verify the mechanical-power balance per unit belt area,

$$\underbrace{(\text{belt-on-liquid shear})\,U}_{\text{belt power}} = \underbrace{\rho g q}_{\text{net rate of potential-energy transport}} + \int_0^h \Phi \,\mathrm{d}y. \qquad [2 \text{ points}]$$

Problem 9: Why the stress is symmetric

[Practice] **This problem is not graded and need not be submitted.** We derived this result in class, so nothing here should be new. Work through it on your own anyway, checking each step rather than reading the argument as a narrative: the identity $\epsilon_{ijk}T_{kj} = 0$ is used without comment throughout the rest of the course, and Problem 1 showed that action and reaction alone will not deliver it. The worked solution is included with this assignment.

Consider a classical nonpolar continuum, with no independent intrinsic angular momentum, no body couples, and no couple stresses. Let O be a fixed origin in an inertial frame and $\boldsymbol{r} = \boldsymbol{x} - \boldsymbol{x}_O$, and let ρ denote the mass density. The angular momentum of a material volume is $\boldsymbol{H}_O = \int_V \rho\, \boldsymbol{r} \times \boldsymbol{v}\, \mathrm{d}V$. You may use the Reynolds transport theorem corollary derived in Lecture 7 and the local linear-momentum balance established in Homework 2,

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_V \rho\psi \,\mathrm{d}V = \int_V \rho \,\frac{\mathrm{D}\psi}{\mathrm{D}t}\,\mathrm{d}V, \qquad \rho\,\frac{\mathrm{D}v_k}{\mathrm{D}t} = \partial_\ell T_{k\ell} + \rho b_k,$$

where $\boldsymbol{b}$ is the body force per unit mass.

(a) First write $\mathrm{d}\boldsymbol{H}_O/\mathrm{d}t$ in terms of the total external torque. Then express the body-force and surface-force torques as integrals, with the moment arm inside each integral.

(b) Show that $\mathrm{D}\left(\boldsymbol{r} \times \boldsymbol{v}\right)/\mathrm{D}t = \boldsymbol{r} \times \mathrm{D}\boldsymbol{v}/\mathrm{D}t$, and use it to simplify the left side of the angular-momentum balance.

(c) Write the balance in indices, substitute $t_k = T_{k\ell} n_\ell$, and apply the divergence theorem to show that

$$\int_V \epsilon_{ijk} r_j \left(\rho\,\frac{\mathrm{D}v_k}{\mathrm{D}t} - \rho b_k - \partial_\ell T_{k\ell}\right)\mathrm{d}V = \int_V \epsilon_{ijk} T_{kj}\,\mathrm{d}V.$$

(d) Use linear-momentum balance and localization to obtain $\epsilon_{ijk}T_{kj} = 0$. Write its three components and deduce the symmetry of the stress. How many independent components remain?

(e) Would the conclusion change if the torque were taken about another fixed origin O'? Explain using the relation between the two angular momenta and the two external torques; do not repeat the entire index calculation.